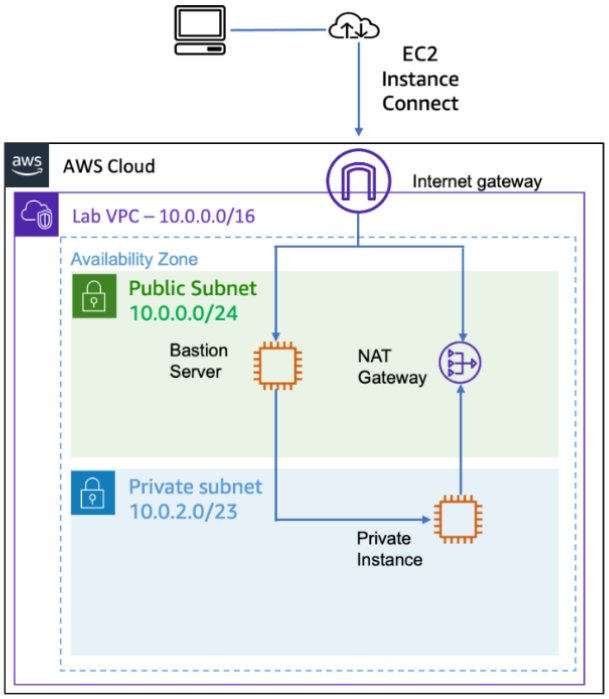
AWS VPC Architecture



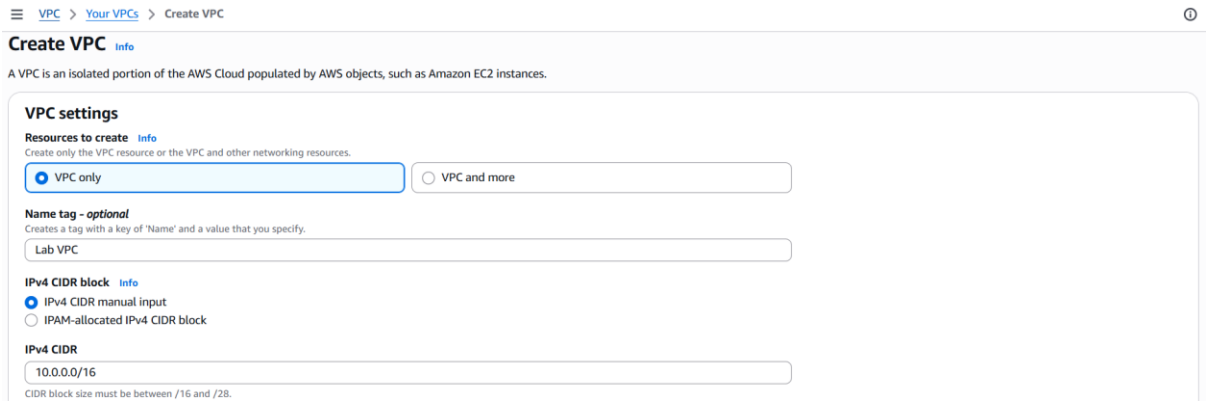
Public Route Table

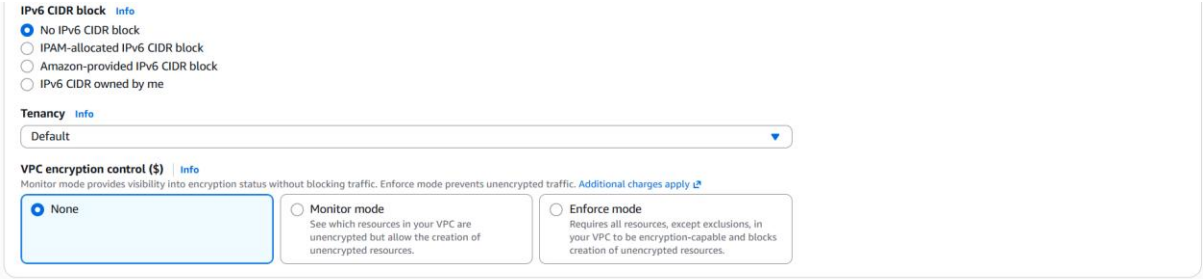
Destination	Target
10.0.0.0/16	Local
0.0.0.0/0	Internet Gateway

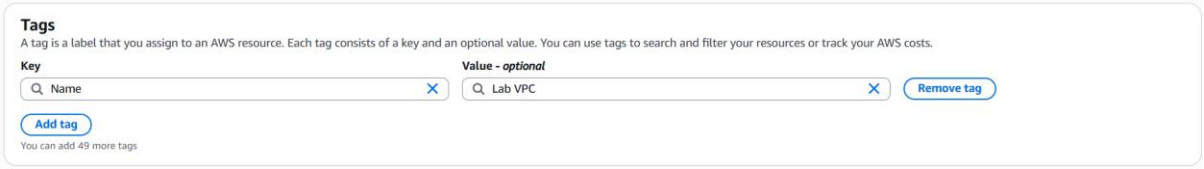
Private Route Table

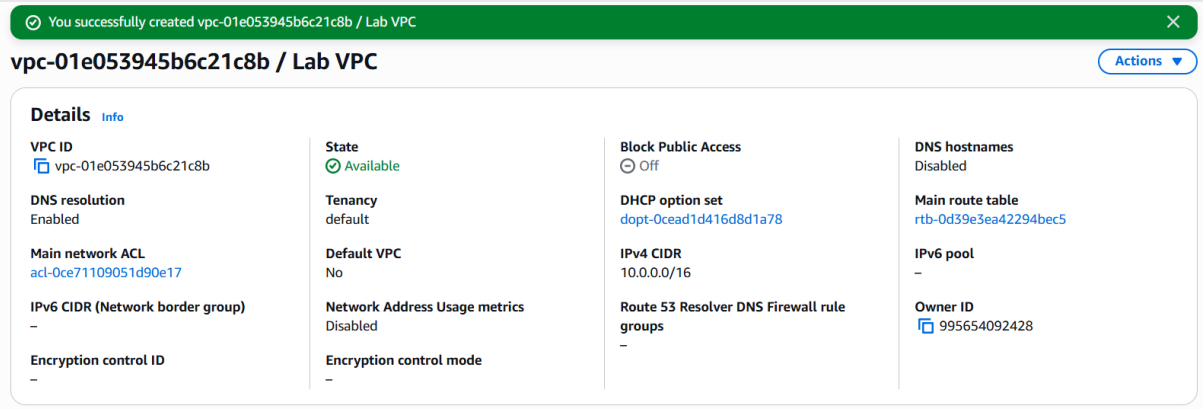
Destination	Target
10.0.0.0/16	Local
0.0.0.0/0	NAT Gateway

Create VPC

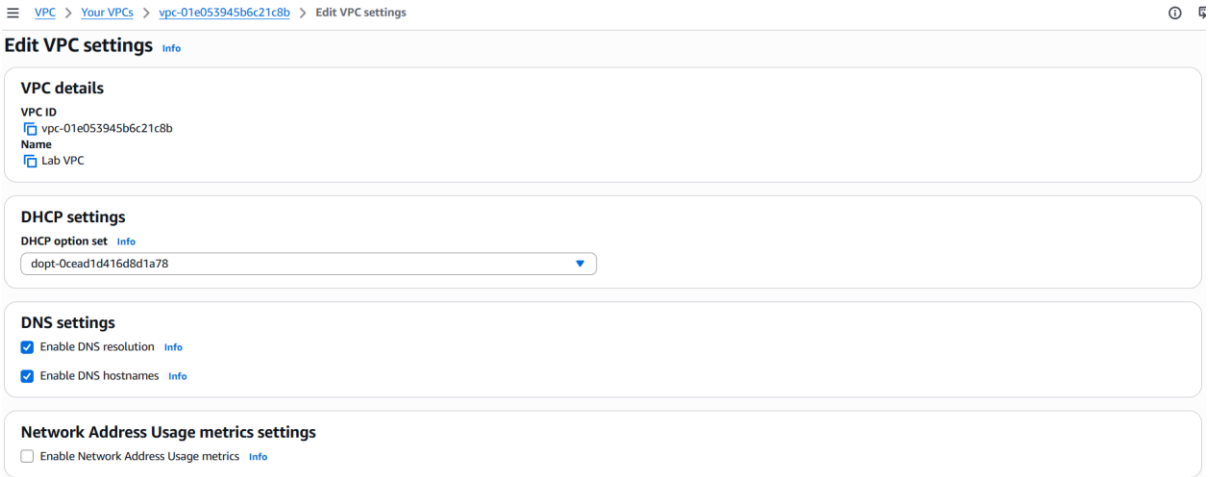


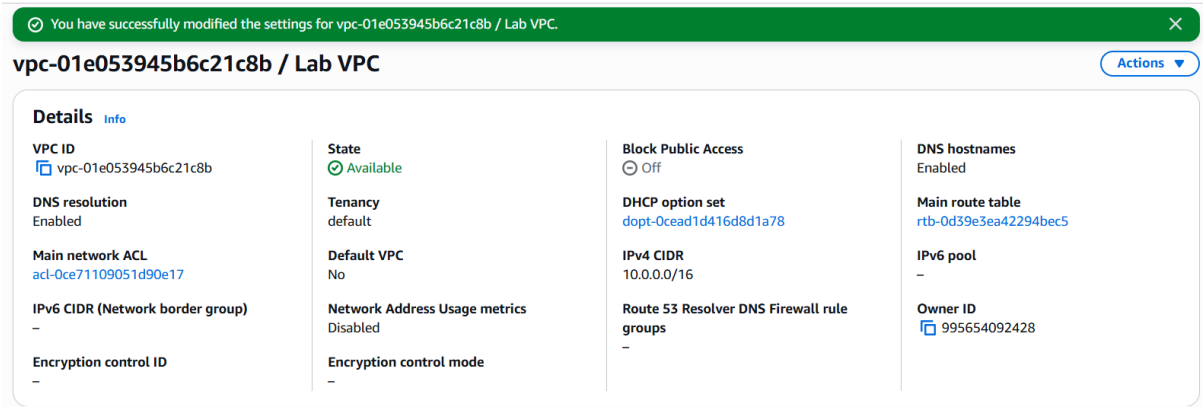






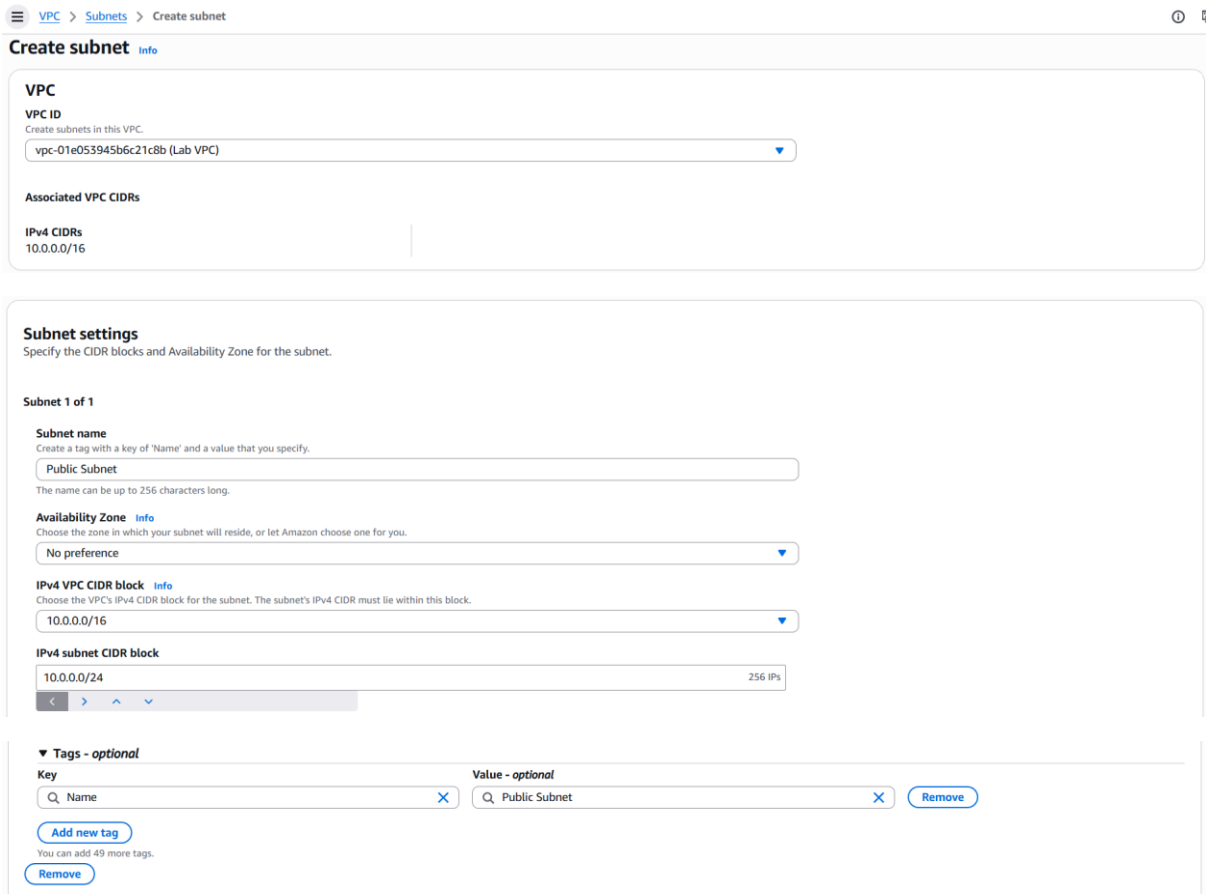
Enable DNS Hostname

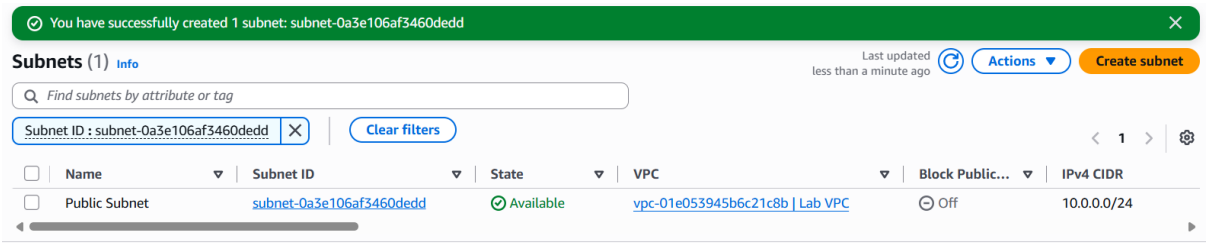




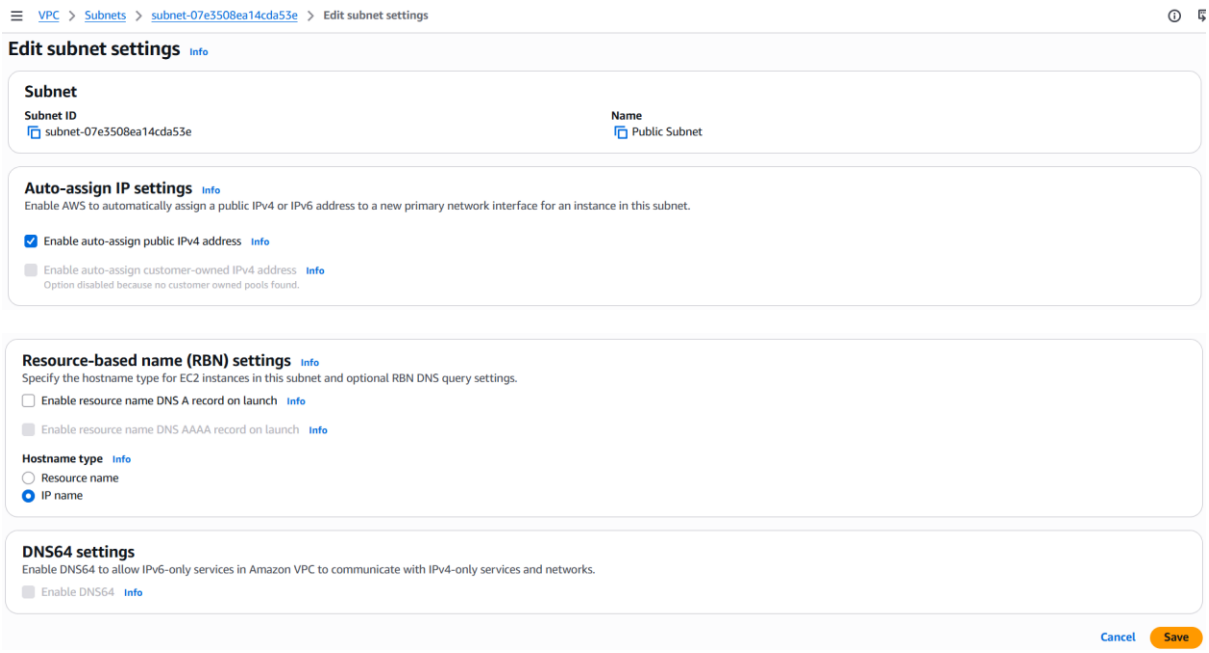
Create Public & Private Subnet

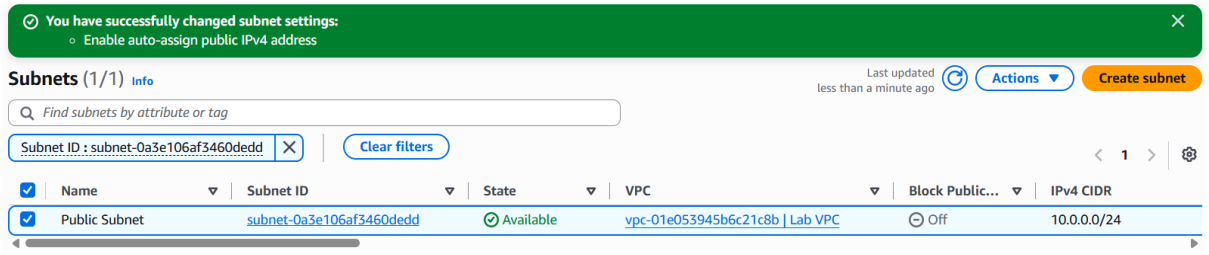
Public Subnet



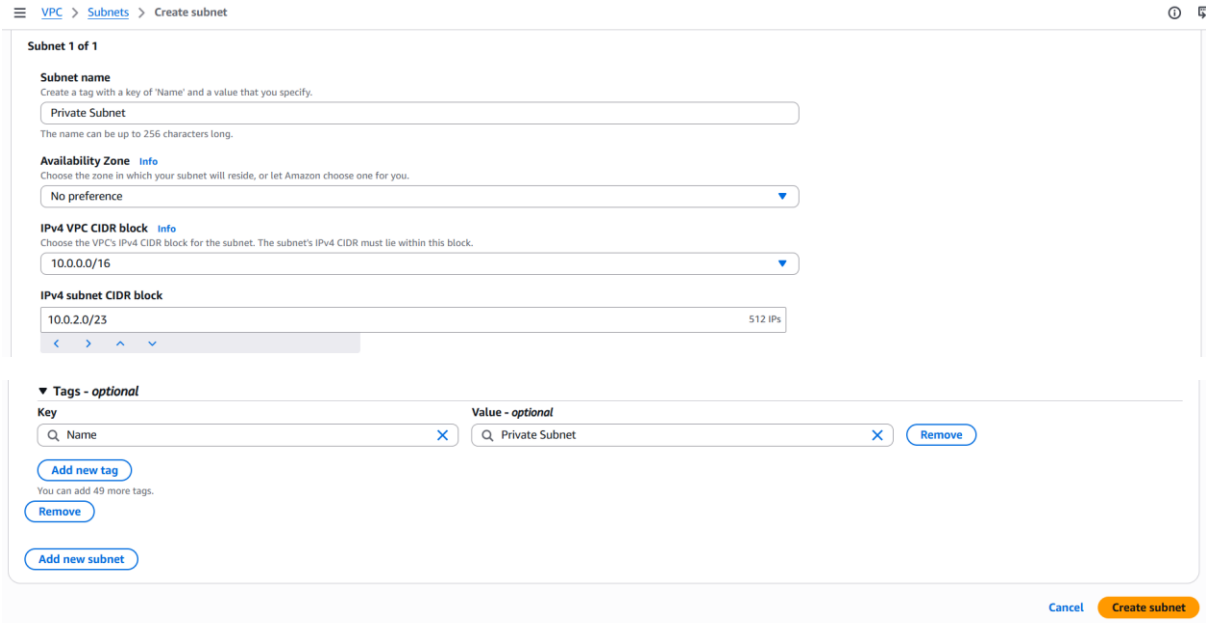


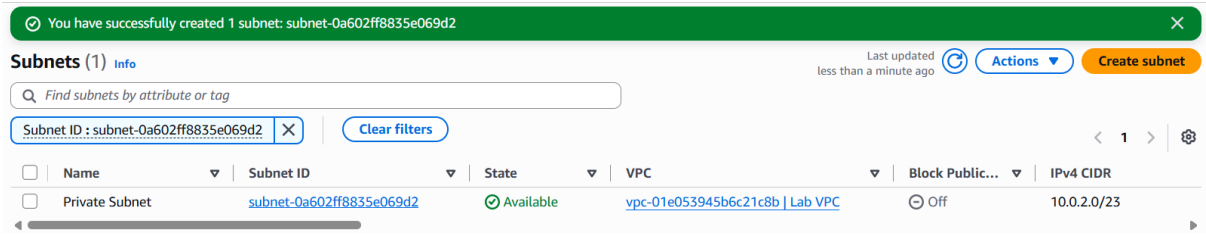
Enable auto-assign public IPv4 address



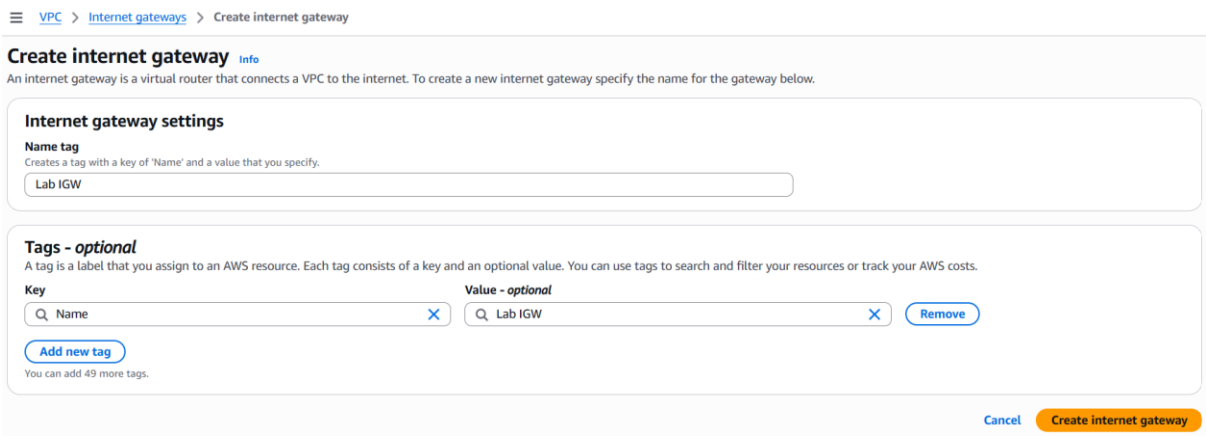


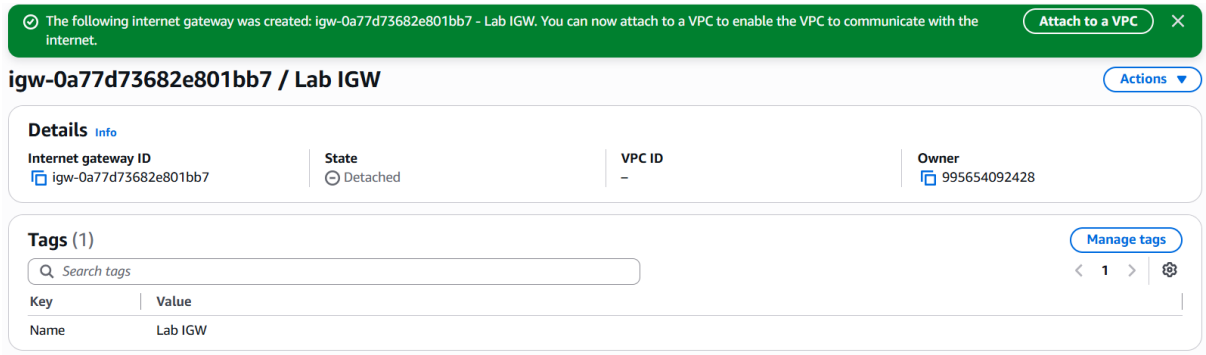
Private Subnet

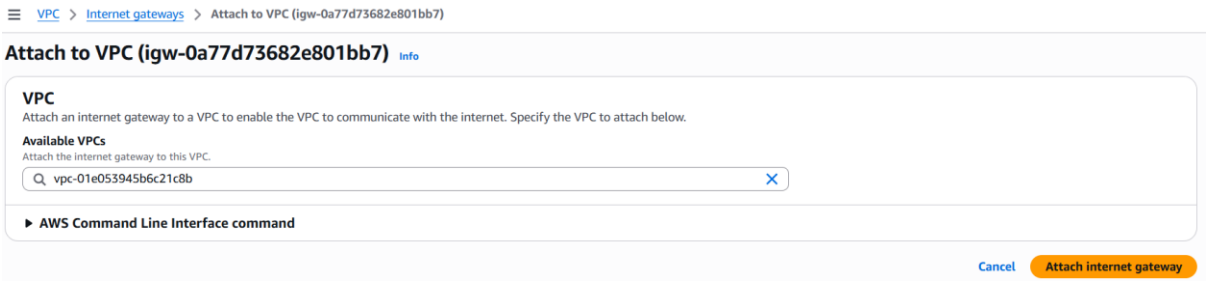


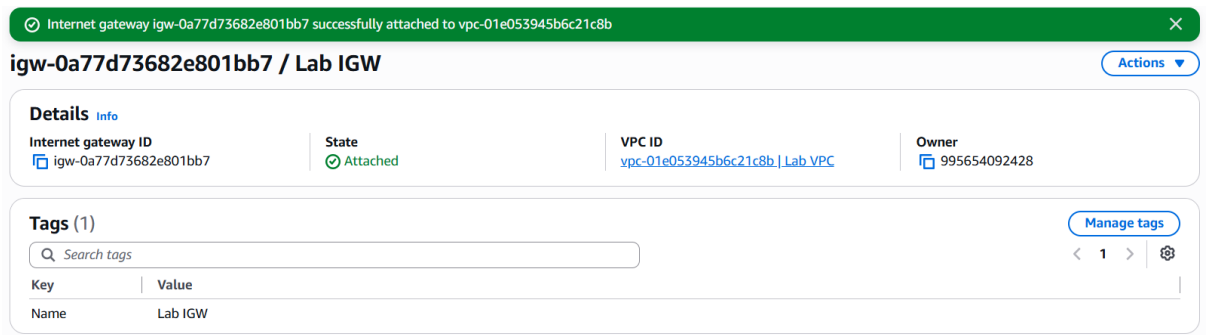


Create an internet gateway

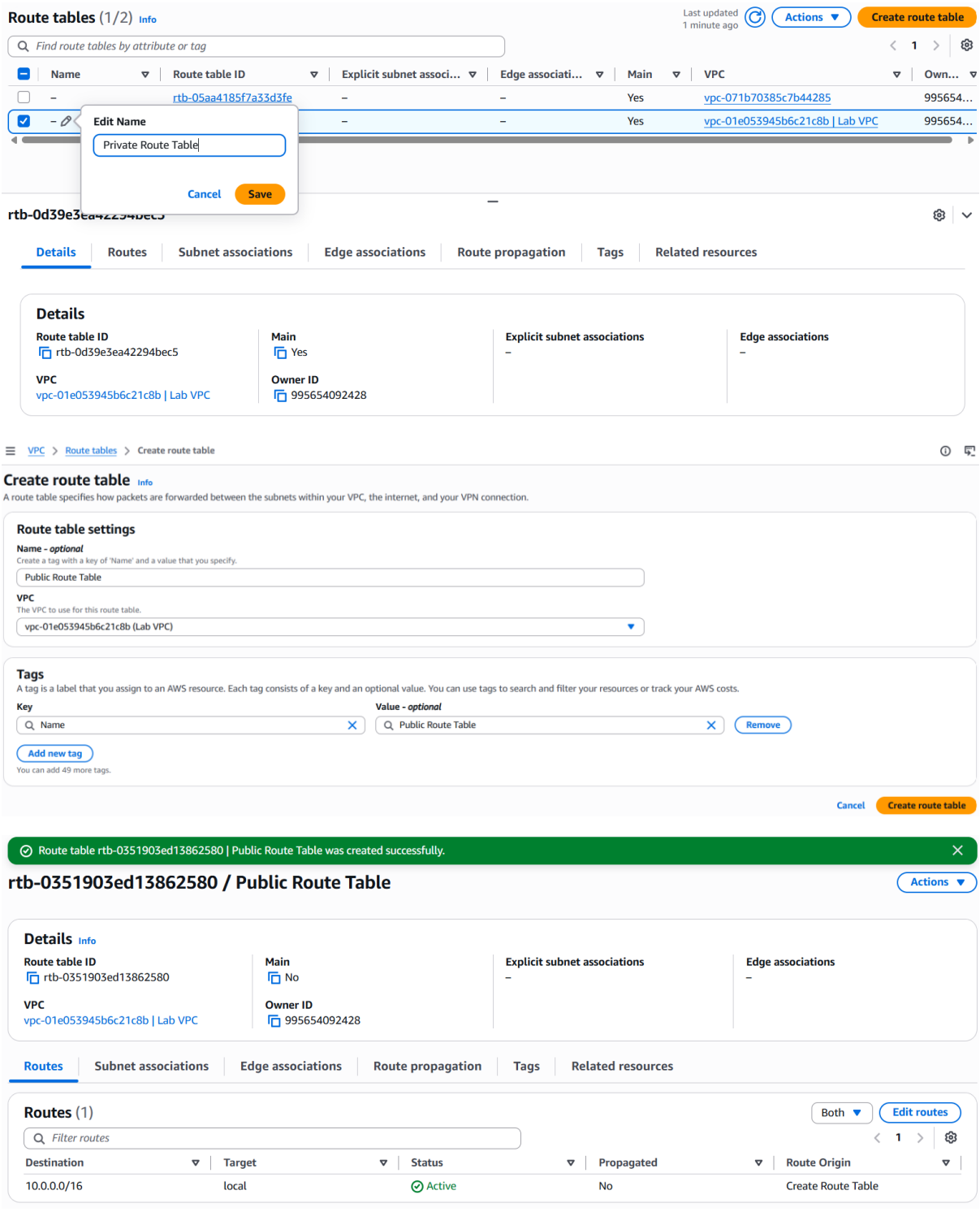


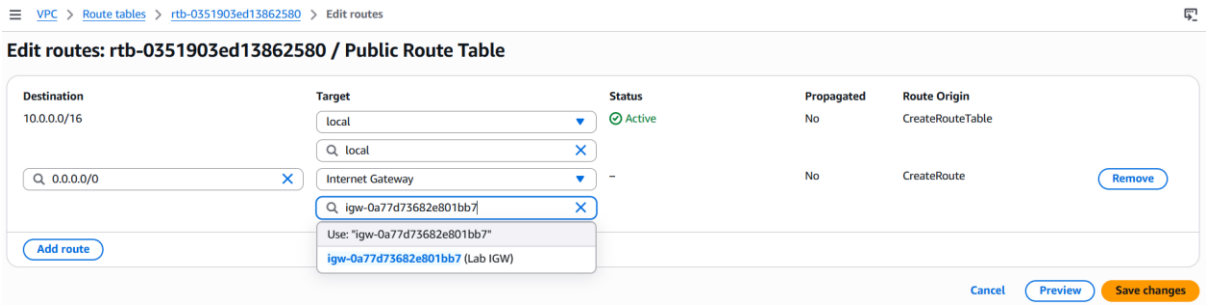


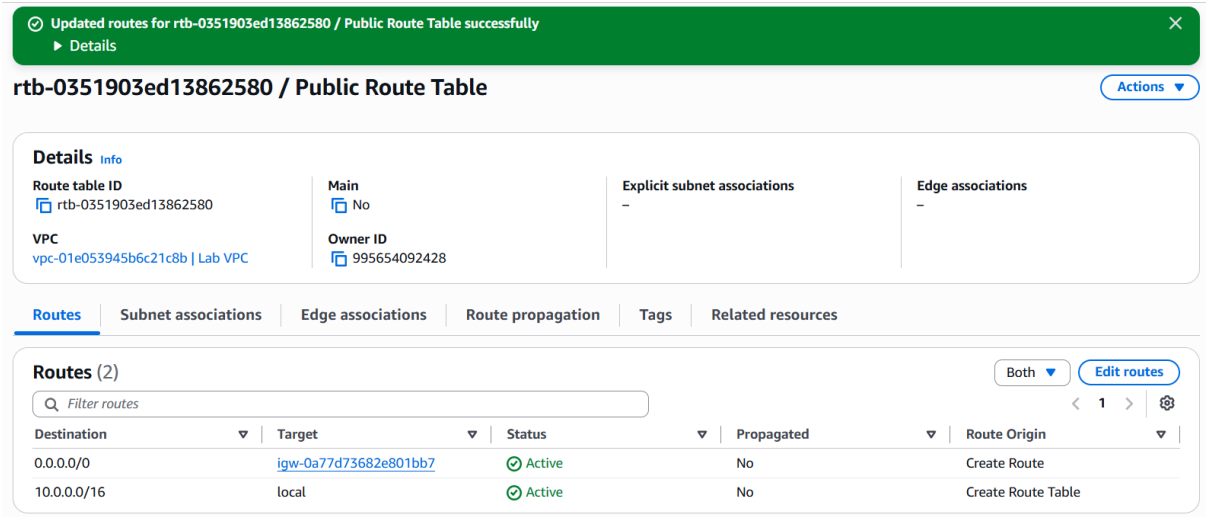




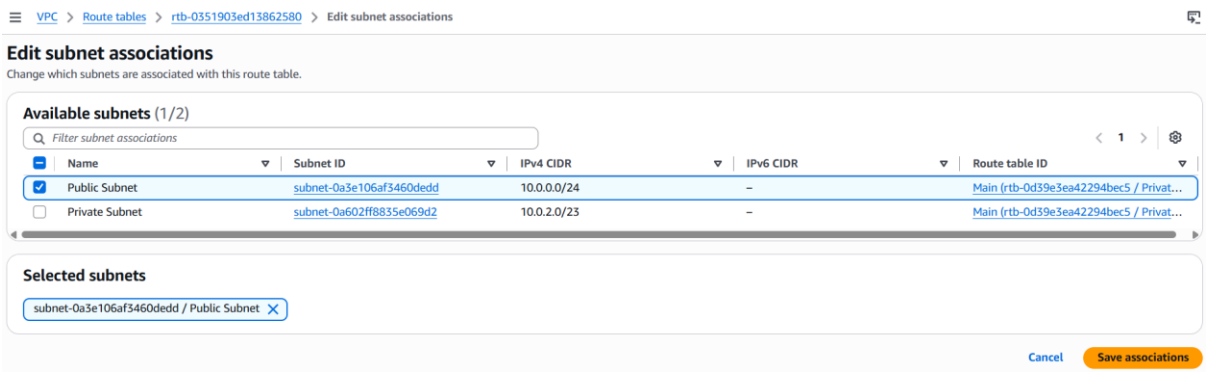
Configure Route Table

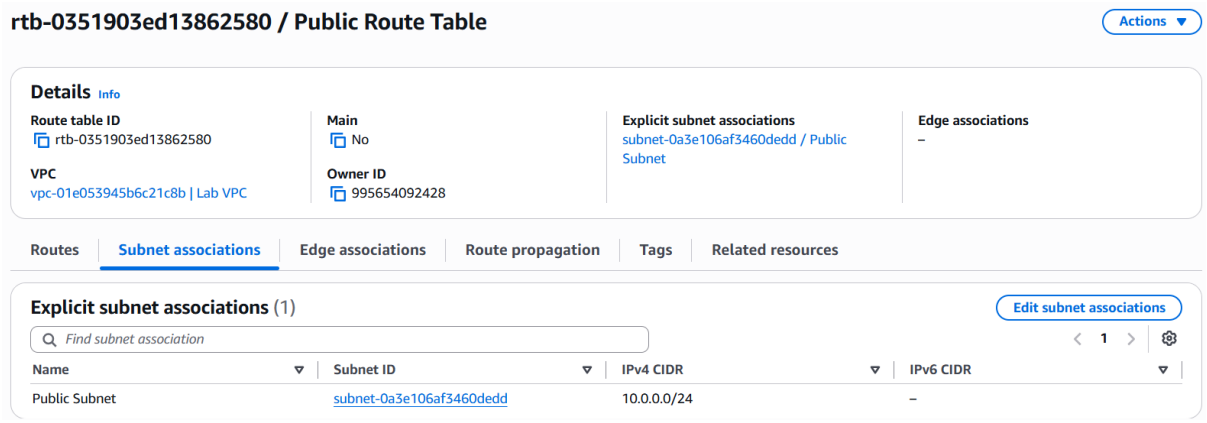




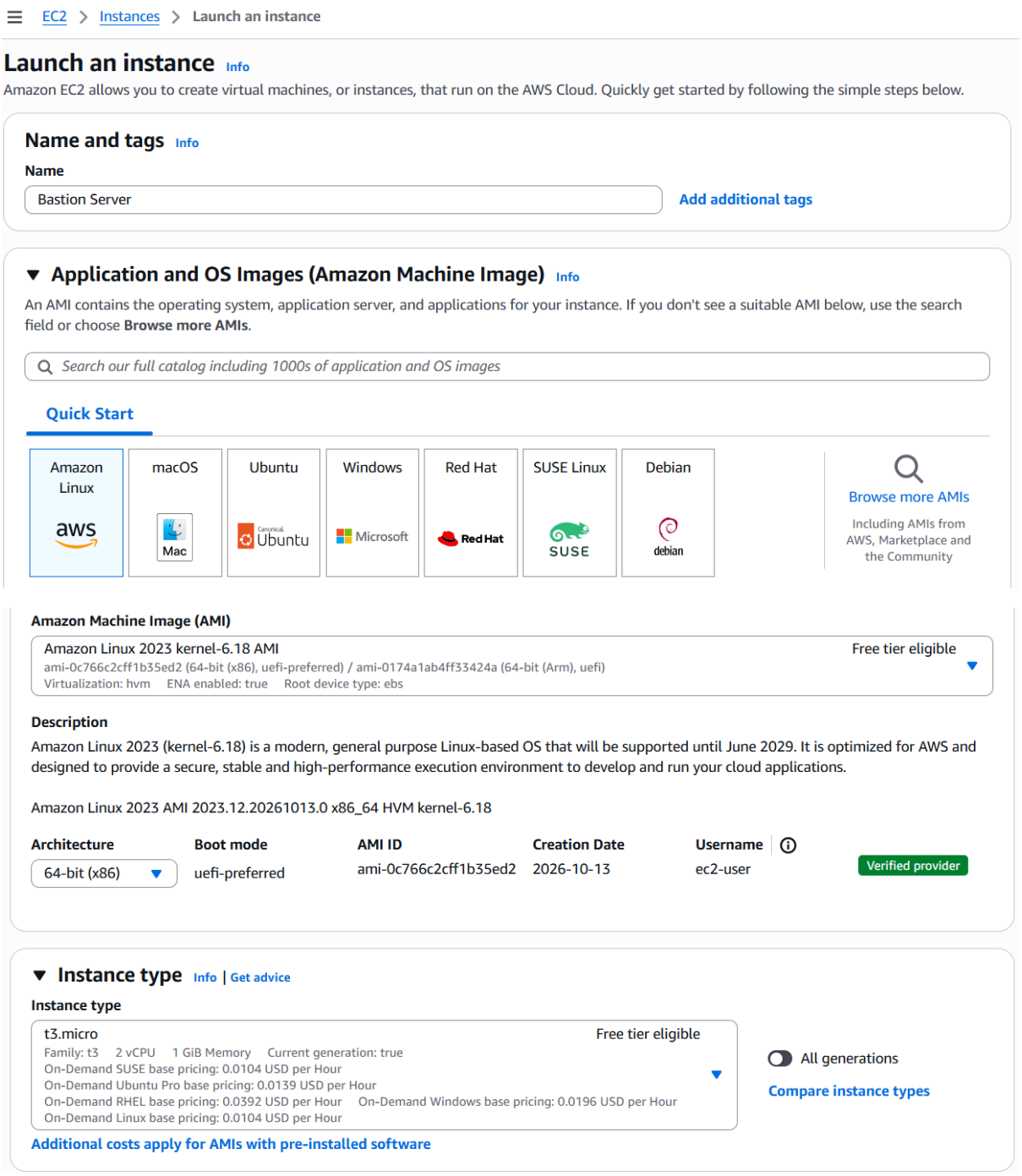


Associate this new route table with the public subnet





Launch bastion server in the public subnet



Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Proceed without a key pair (Not recommended) Default value

Create new key pair

Network settings Info

VPC - *required* Info

vpc-01e053945b6c21c8b (Lab VPC)
10.0.0.0/16

Subnet Info

subnet-0a3e106af3460dedd Public Subnet
VPC: vpc-01e053945b6c21c8b Owner: 995654092428 Availability Zone: us-west-2a (usw2-az2)
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.0.0/24)

Create new subnet

Auto-assign public IP Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- (•) Create security group
- ( ) Select existing security group

Security group name - *required*

Bastion Security Group

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!$*

Description - *required* Info

Allow SSH

Inbound Security Group Rules

Security group rule 1 (TCP, 22, 0.0.0.0/0) Remove

Type Info	Protocol Info	Port range Info
ssh	TCP	22

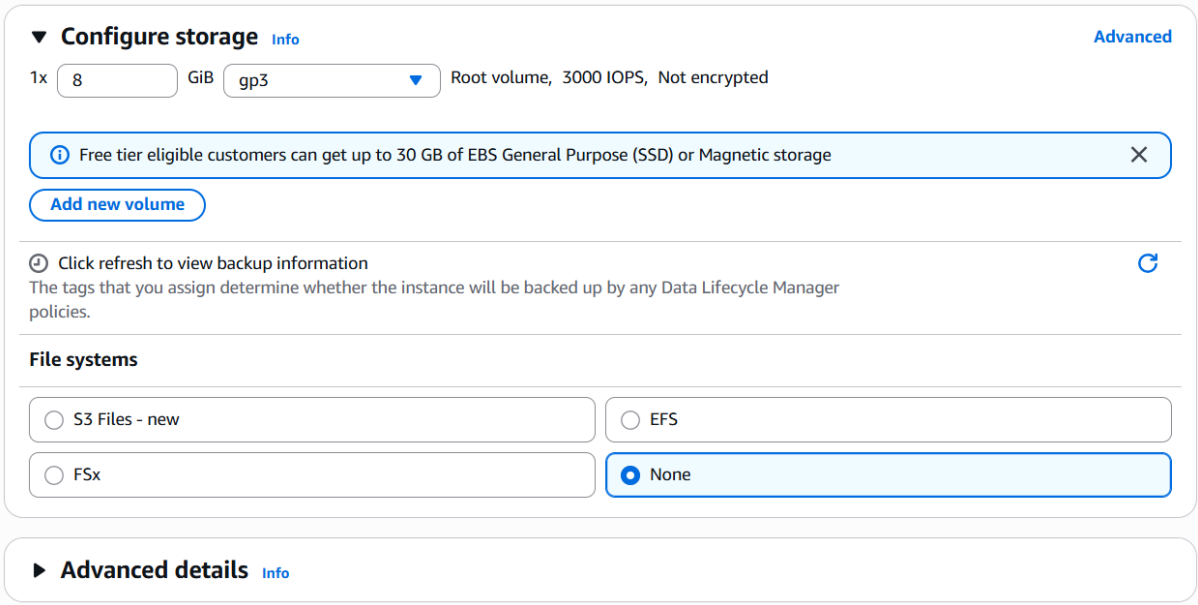
Source type Info	Source Info	Description - *optional* Info
Anywhere	Add CIDR, prefix list or security group	e.g. SSH for admin desktop

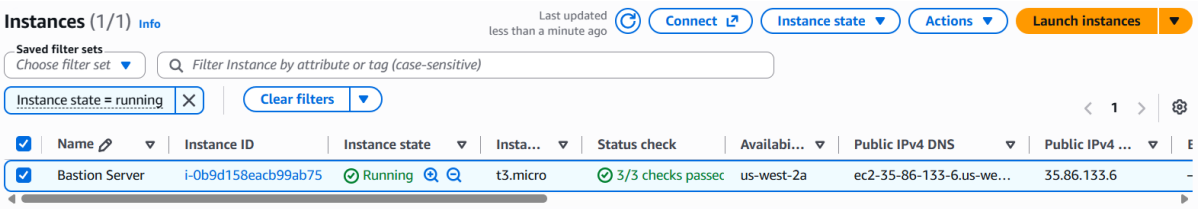
0.0.0.0/0

⚠ Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

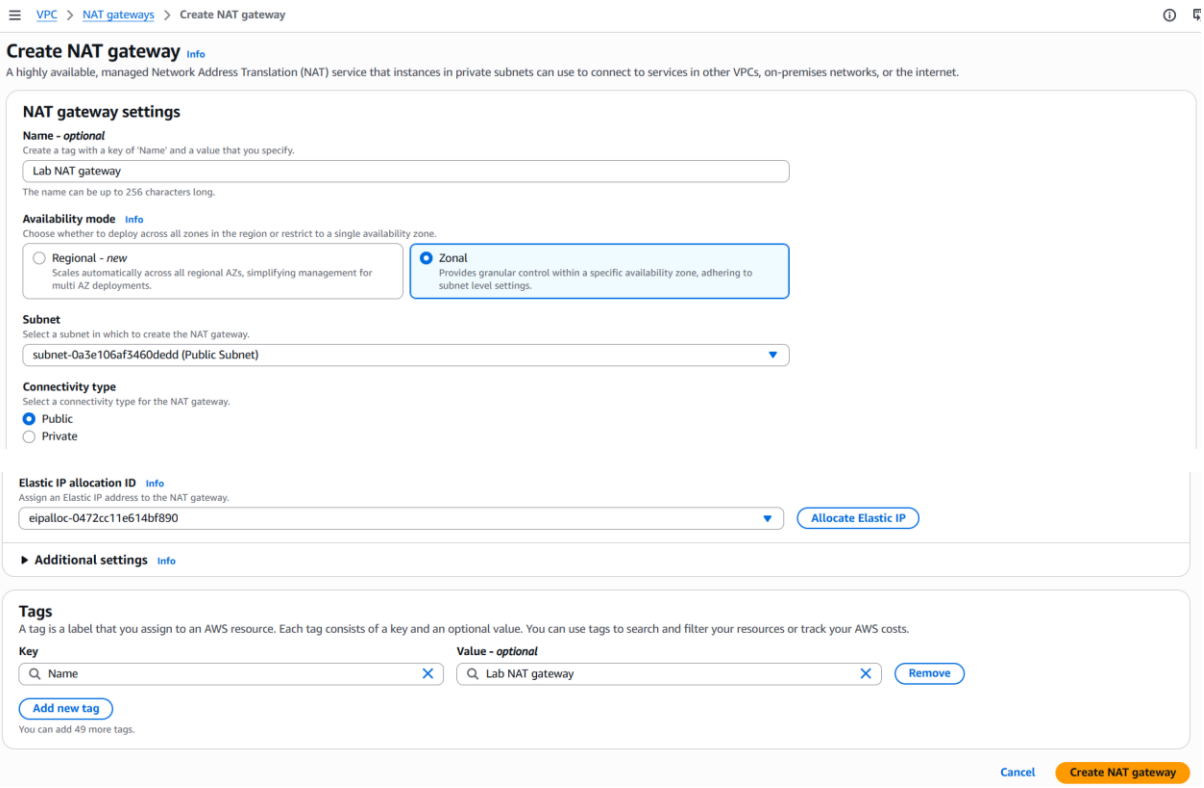
Add security group rule

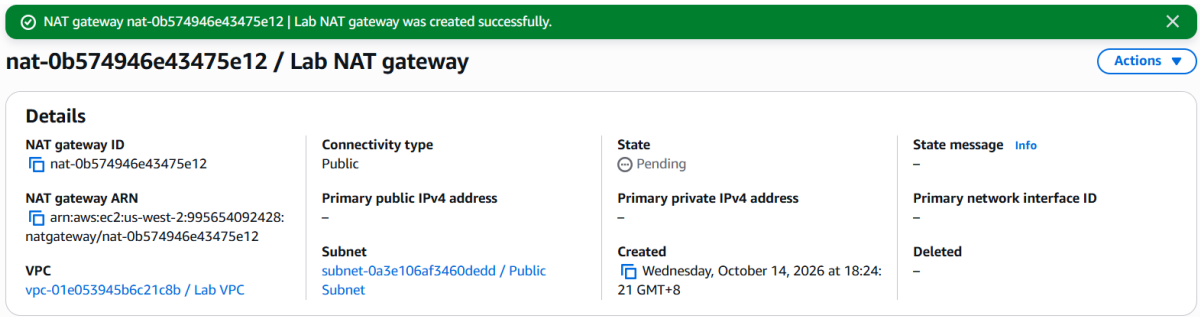
▶ **Advanced network configuration**

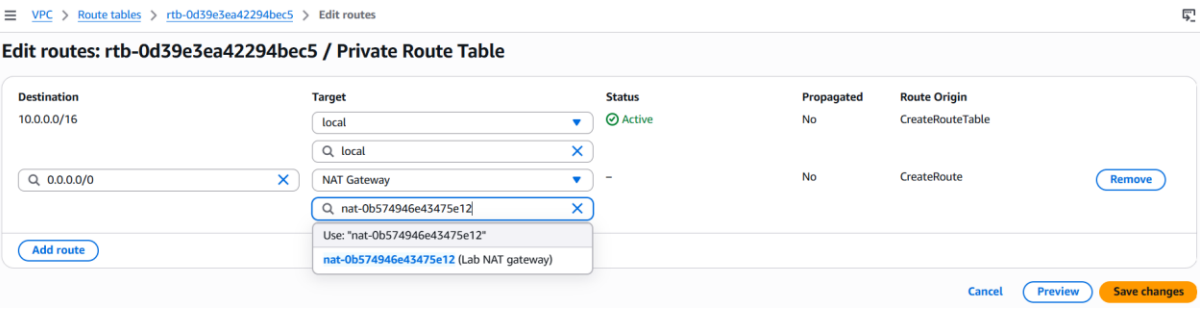


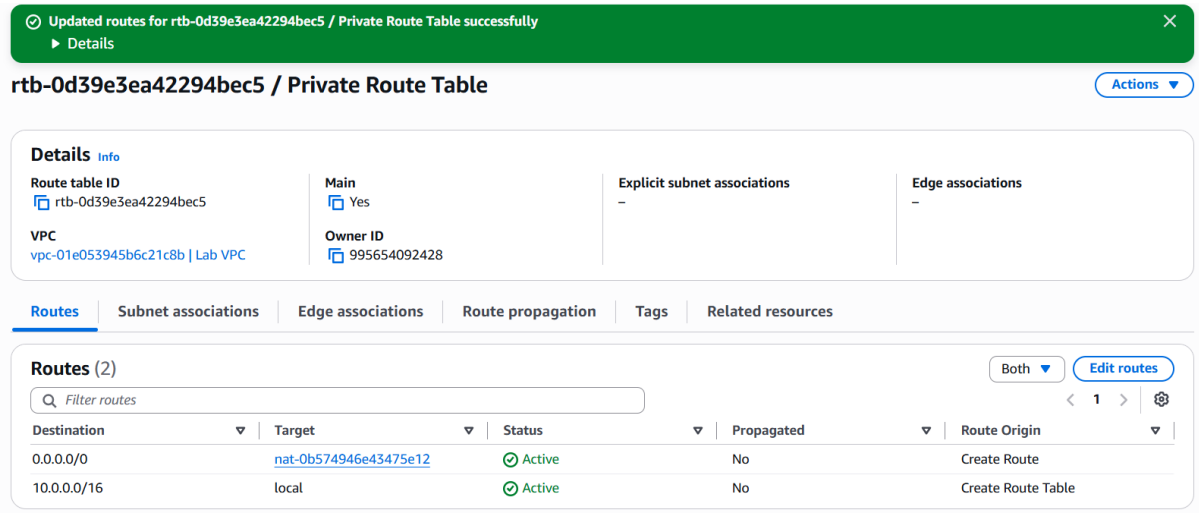


Create a NAT gateway

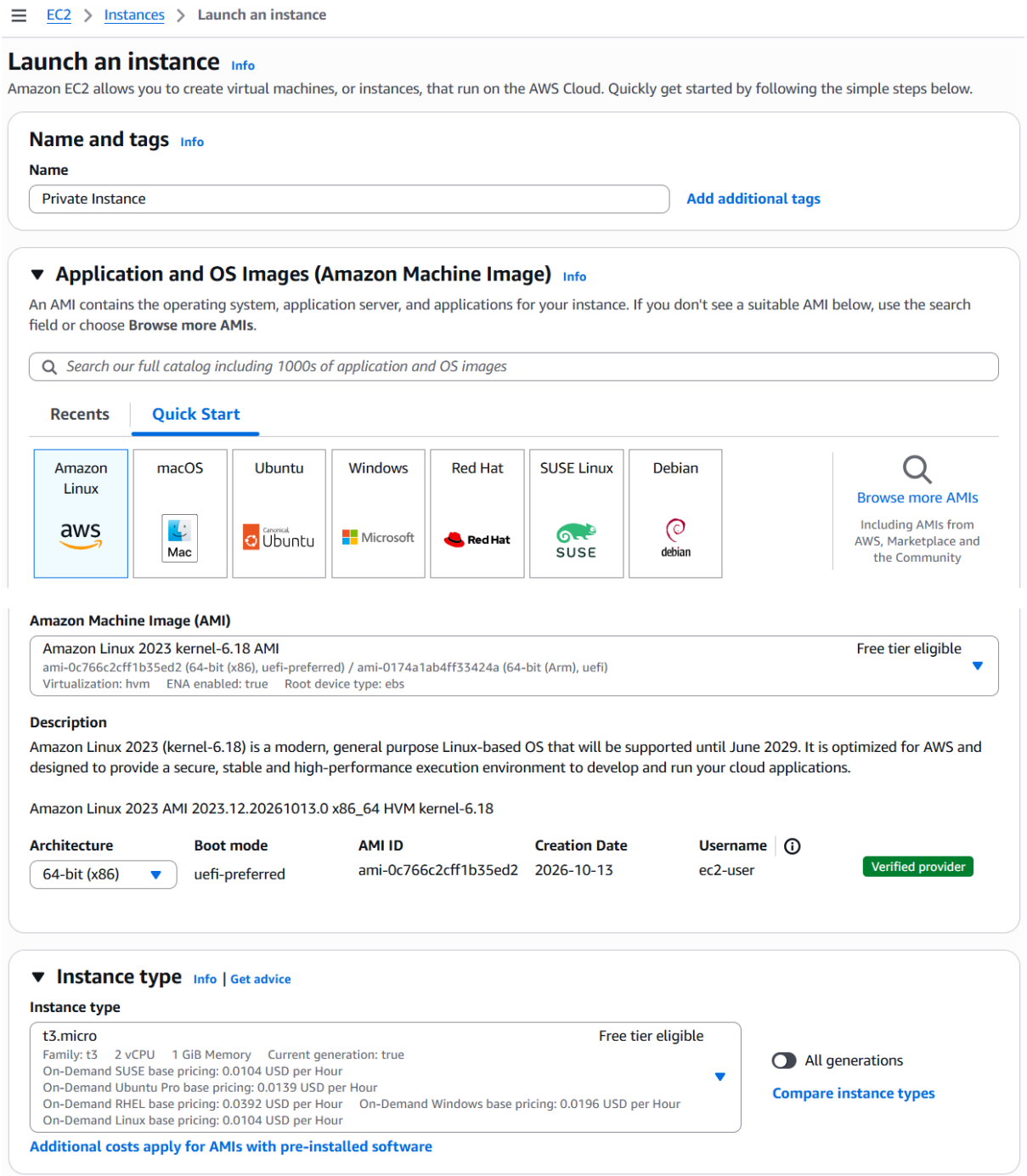



NAT gateway nat-0b574946e43475e12 | Lab NAT gateway was created successfully.
nat-0b574946e43475e12 / Lab NAT gateway
Actions
Details
NAT gateway ID
nat-0b574946e43475e12
NAT gateway ARN
arn:aws:ec2:us-west-2:995654092428:natgateway/nat-0b574946e43475e12
VPC
vpc-01e053945b6c21c8b / Lab VPC
Connectivity type
Public
Primary public IPv4 address
–
Subnet
subnet-0a3e106af3460dedd / Public Subnet
State
Pending
Primary private IPv4 address
–
Created
Wednesday, October 14, 2026 at 18:24:21 GMT+8
State message
Info
–
Primary network interface ID
–
Deleted
–


VPC > Route tables > rtb-0d39e3ea42294bec5 > Edit routes
Edit routes: rtb-0d39e3ea42294bec5 / Private Route Table
Destination
Target
Status
Propagated
Route Origin
10.0.0.0/16
local
Active
No
CreateRouteTable
local
0.0.0.0/0
NAT Gateway
–
No
CreateRoute
Remove
nat-0b574946e43475e12
Use: "nat-0b574946e43475e12"
nat-0b574946e43475e12 (Lab NAT gateway)
Add route
Cancel
Preview
Save changes


Updated routes for rtb-0d39e3ea42294bec5 / Private Route Table successfully
Details
rtb-0d39e3ea42294bec5 / Private Route Table
Actions
Details
Info
Route table ID
rtb-0d39e3ea42294bec5
VPC
vpc-01e053945b6c21c8b | Lab VPC
Main
Yes
Owner ID
995654092428
Explicit subnet associations
–
Edge associations
–
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Routes (2)
Both
Edit routes
Filter routes
1
Destination
Target
Status
Propagated
Route Origin
0.0.0.0/0
nat-0b574946e43475e12
Active
No
Create Route
10.0.0.0/16
local
Active
No
Create Route Table

Launch an instance in the private subnet



Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

Proceed without a key pair (Not recommended) — Default value

Create new key pair

Network settings Info

VPC - *required* Info

vpc-01e053945b6c21c8b (Lab VPC)
10.0.0.0/16

Subnet Info

subnet-0a602ff8835e069d2 — Private Subnet
VPC: vpc-01e053945b6c21c8b Owner: 995654092428 Availability Zone: us-west-2a (usw2-az2)
Zone type: Availability Zone IP addresses available: 507 CIDR: 10.0.2.0/23)

Create new subnet

Auto-assign public IP Info

Disable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

- (•) Create security group
- ( ) Select existing security group

Security group name - *required*

Private Instance SG

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!$*

Description - *required* Info

Allow SSH from Bastion

Inbound Security Group Rules

Security group rule 1 (TCP, 22, 10.0.0.0/16) — Remove

Type Info	Protocol Info	Port range Info
ssh	TCP	22

Source type Info	Source Info	Description - *optional* Info
Custom	Add CIDR, prefix list or security group	e.g. SSH for admin desktop

10.0.0.0/16

Add security group rule

▶ **Advanced network configuration**

Configure storage Info

Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

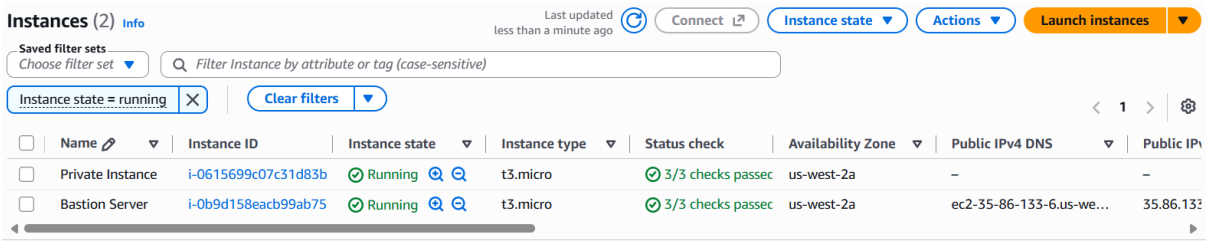
Add new volume

Click refresh to view backup information

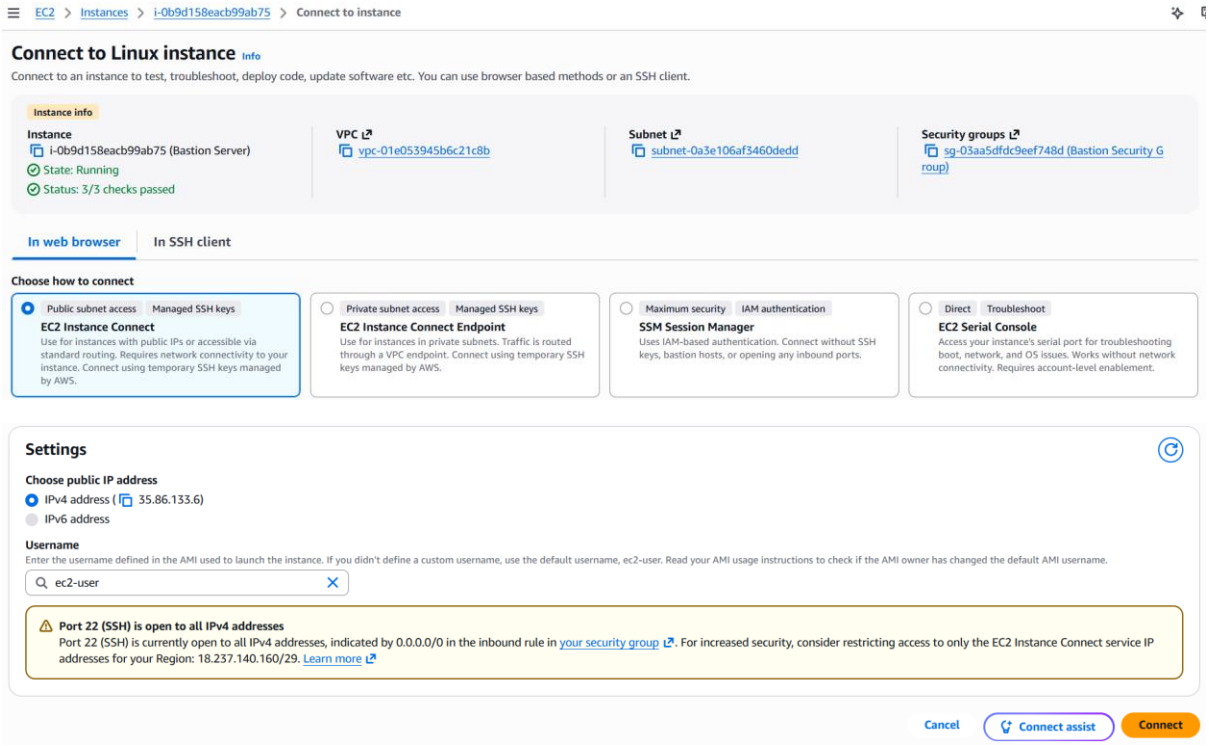
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

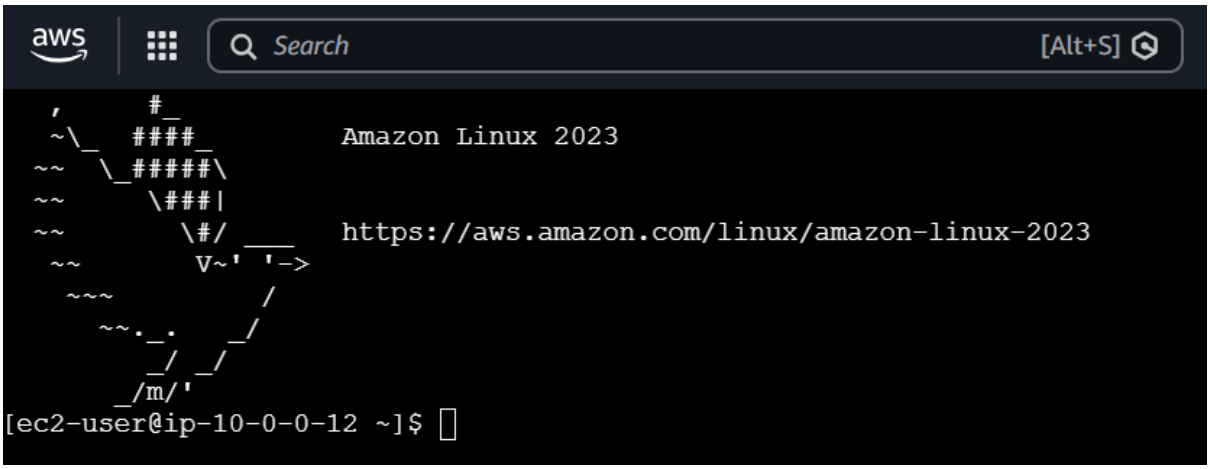
File systems

- ( ) S3 Files - new
- ( ) EFS
- ( ) FSx
- (•) None

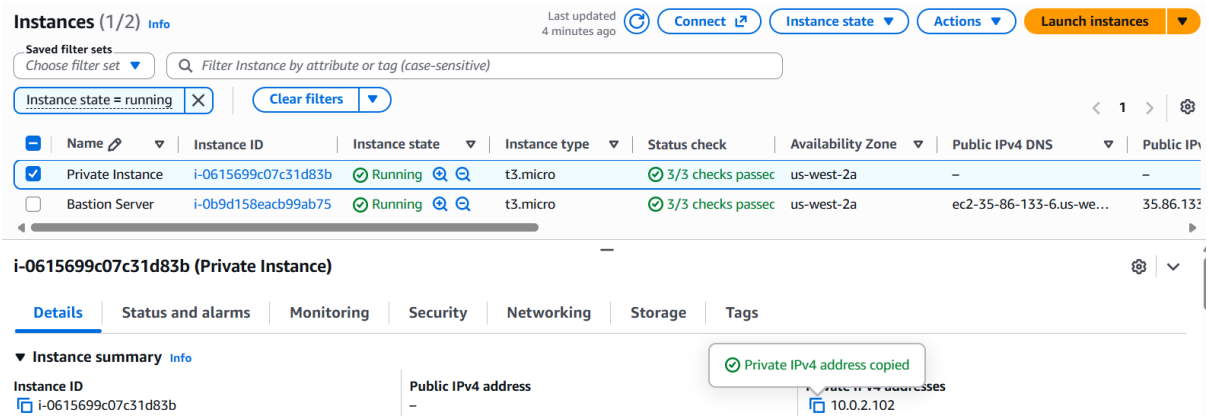


Logging in to the bastion server



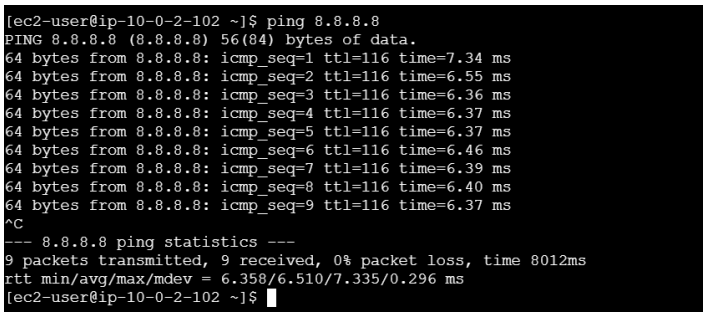


Logging in to the private instance





Testing the NAT gateway



This output indicates that the private instance successfully communicated with google.com on the internet. The private instance is in the private subnet, and the only way that this is possible in the current scenario is by going through the NAT gateway.